

Modular Arithmetic

Objectives

- ❑ To emphasize the importance of **modular arithmetic** and the **modulo operator**, because they are extensively used in cryptography
- ❑ To emphasize and review matrices and operations on residue matrices that are extensively used in cryptography

Modular Arithmetic

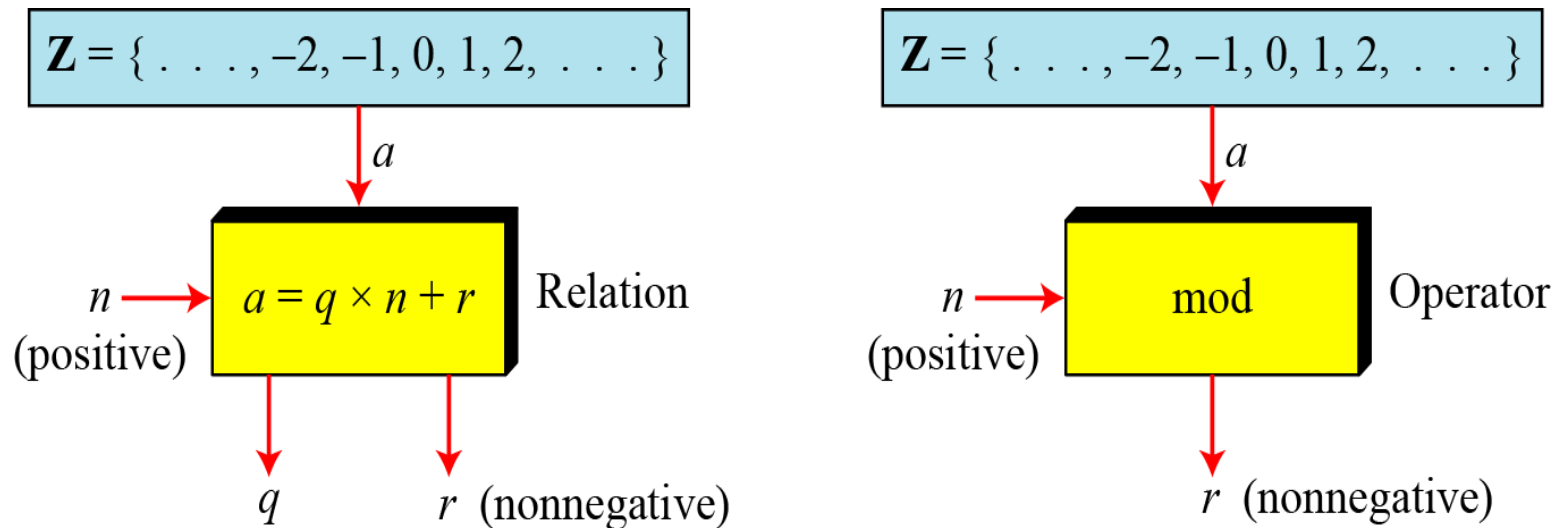
- Several important cryptosystems make use of modular arithmetic.
- This is when the answer to a calculation is always in the range $0 - m$ where m is the **modulus**.

Modular Arithmetic

- The division relationship ($a = q \times n + r$) discussed in the previous section has two inputs (a and n) and two outputs (q and r).
- In modular arithmetic, we are interested in only one of the outputs, the remainder r .

Modulo Operator

- The modulo operator is shown as **mod**. The second input (n) is called the modulus. The output r is called the residue.



Division algorithm and modulo operator

Example

Find the result of the following operations:

a. $27 \bmod 5$

b. $36 \bmod 12$

c. $-18 \bmod 14$

d. $-7 \bmod 10$

Solution

a. Dividing 27 by 5 results in $r = 2$

b. Dividing 36 by 12 results in $r = 0$.

c. Dividing -18 by 14 results in $r = -4$. After adding the modulus $r = 10$

d. Dividing -7 by 10 results in $r = -7$. After adding the modulus to -7 , $r = 3$.

Divisibility

- If a is not zero and we let $r = 0$ in the division relation, we get $a = q \times n$

n divides a (or n is a divisor of a) $a \mid n$

If the remainder is not zero, then n does not divide

Example

- a. The integer 4 divides the integer 32 because $32 = 8 \times 4$. We show this as $4|32$.
- b. The number 8 does not divide the number 42 because $42 = 5 \times 8 + 2$. There is a remainder, the number 2, in the equation. We show this as $8 \nmid 42$.

Properties of divisibility

Property 1: if $a|1$, then $a = \pm 1$.

Property 2: if $a|b$ and $b|a$, then $a = \pm b$.

Property 3: if $a|b$ and $b|c$, then $a|c$.

Property 4: if $a|b$ and $a|c$, then $a|(m \times b + n \times c)$, where m and n are arbitrary integers.

Example

- a. Since $3|15$ and $15|45$, according to the third property, $3|45$.
- b. Since $3|15$ and $3|9$, according to the fourth property, $3|(15 \times 2 + 9 \times 4)$, which means $3|66$.

All Divisors

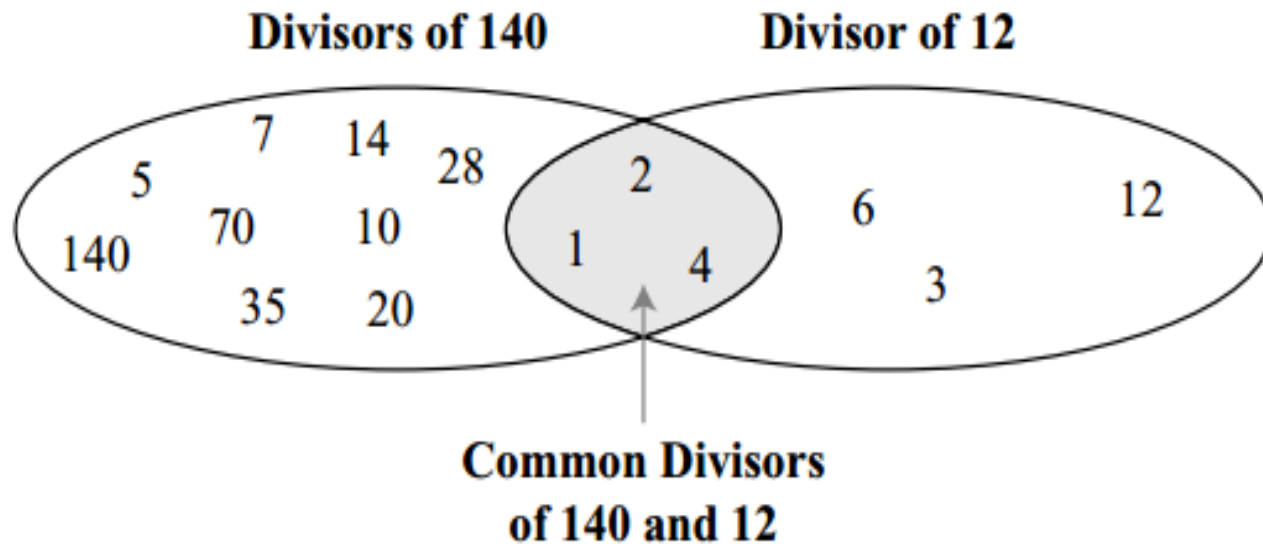
- A positive integer can have more than one divisor

Fact 1: The integer 1 has only one divisor, itself.

Fact 2: Any positive integer has at least two divisors, 1 and itself (but it can have more).

Greatest Common Divisor

- Two positive integers may have many common divisors, but only one greatest common divisor.



Euclidean Algorithm

- It can find the greatest common divisor of two positive integers.
- Based on the following two facts.

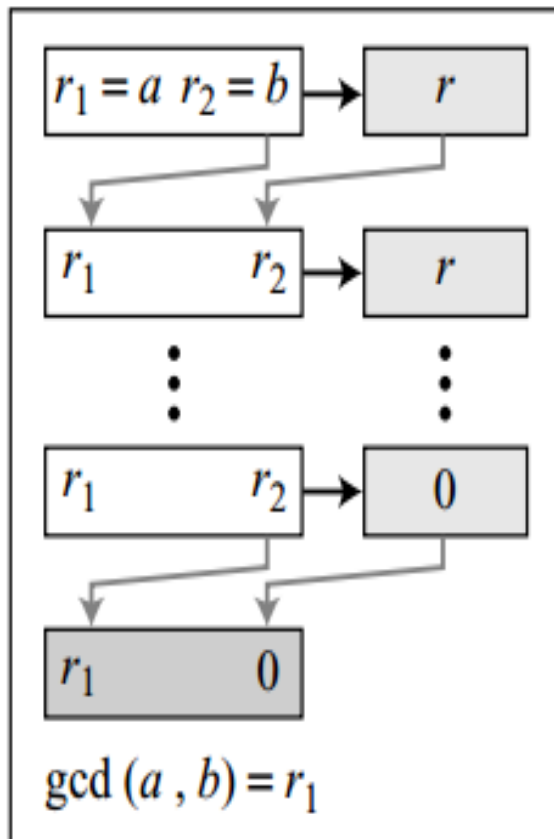
Fact 1: $\gcd(a, 0) = a$

Fact 2: $\gcd(a, b) = \gcd(b, r)$, where r is the remainder of dividing a by b

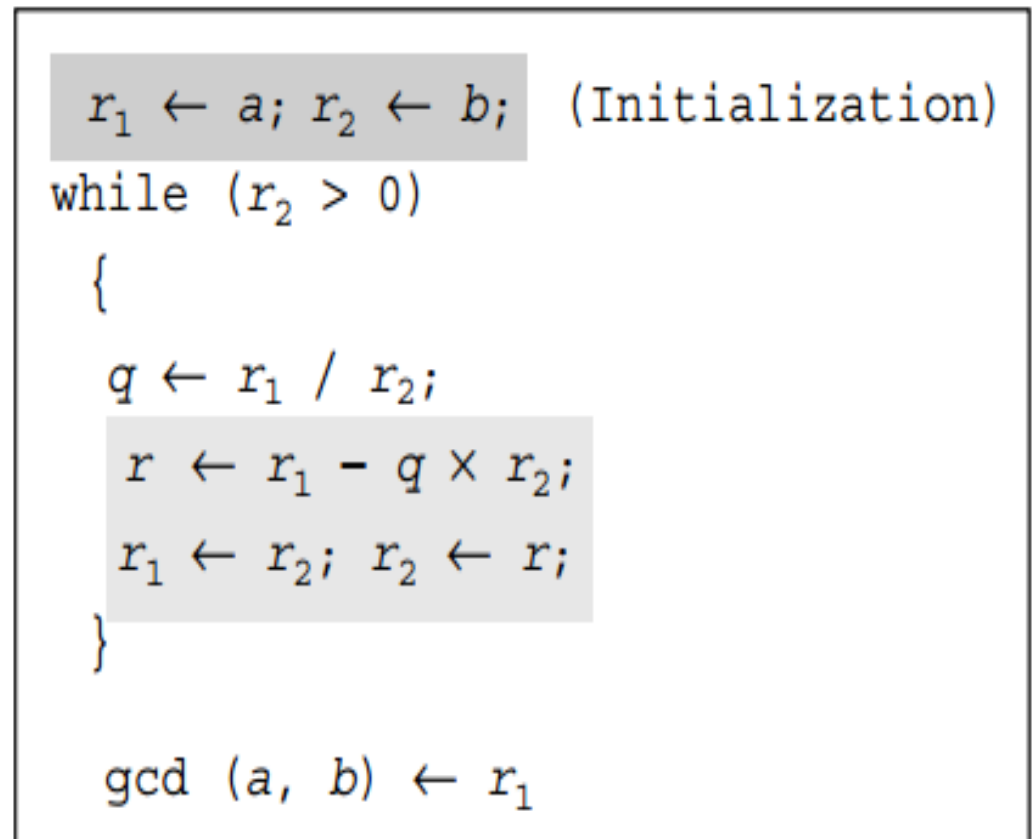
$$\gcd(36, 10) = \gcd(10, 6) = \gcd(6, 4) = \gcd(4, 2) = \gcd(2, 0) = 2$$

When $\gcd(a, b) = 1$, we say that a and b are relatively prime.

Euclidean algorithm



a. Process



b. Algorithm

Example

- Find the greatest common divisor of 2740 and 1760

q	r_1	r_2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	20	0	

Example

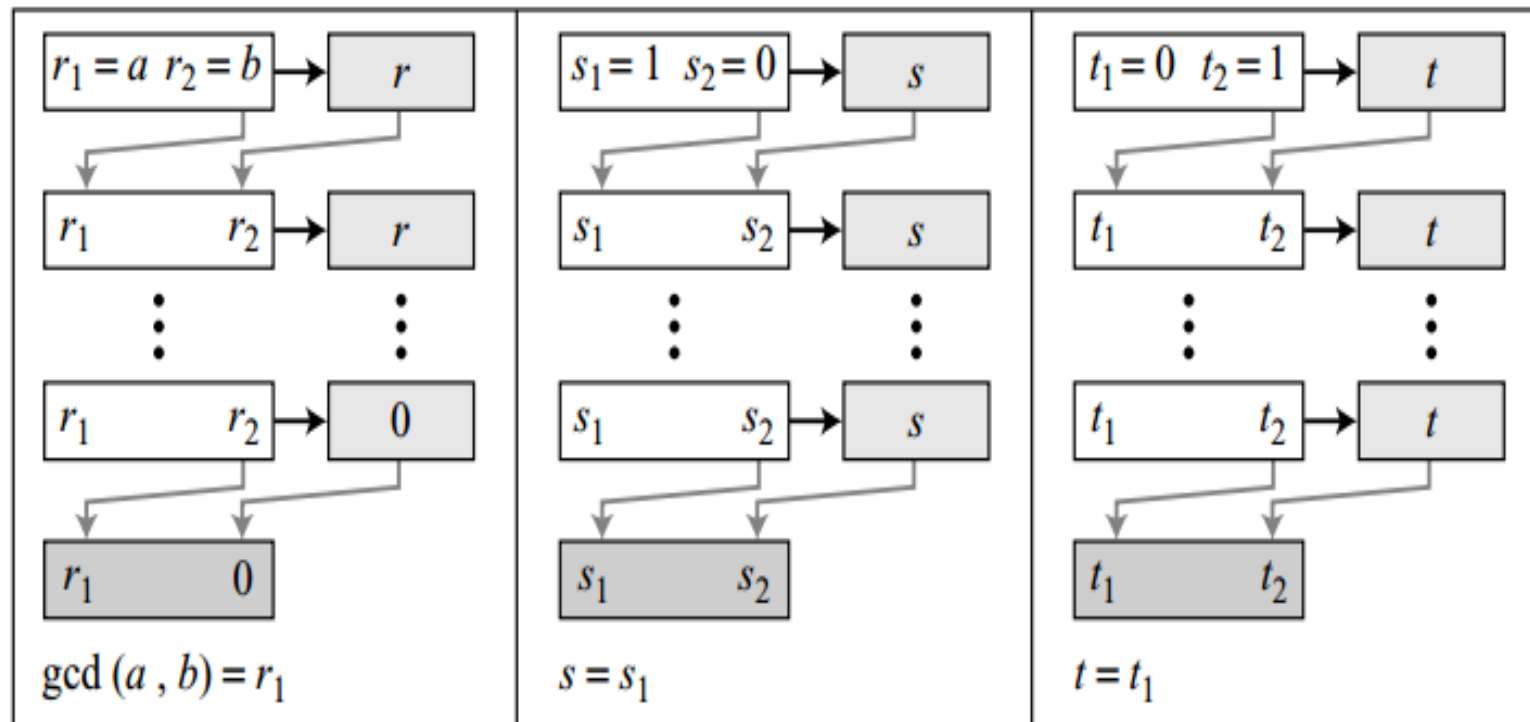
- Find the greatest common divisor of 25 and 60

q	r_1	r_2	r
0	25	60	25
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

Extended Euclidean Algorithm

- Given two integers a and b , we often need to find other two integers, s and t , such that
$$s \times a + t \times b = \gcd(a, b)$$
-

Extended Euclidean algorithm



a. Process

Extended Euclidean algorithm

```
 $r_1 \leftarrow a; r_2 \leftarrow b;$   
 $s_1 \leftarrow 1; s_2 \leftarrow 0;$  (Initialization)  
 $t_1 \leftarrow 0; t_2 \leftarrow 1;$   
while ( $r_2 > 0$ )  
{  
   $q \leftarrow r_1 / r_2;$   
   $r \leftarrow r_1 - q \times r_2;$  (Updating  $r$ 's)  
   $r_1 \leftarrow r_2; r_2 \leftarrow r;$   
   $s \leftarrow s_1 - q \times s_2;$  (Updating  $s$ 's)  
   $s_1 \leftarrow s_2; s_2 \leftarrow s;$   
   $t \leftarrow t_1 - q \times t_2;$  (Updating  $t$ 's)  
   $t_1 \leftarrow t_2; t_2 \leftarrow t;$   
}  
gcd ( $a, b$ )  $\leftarrow r_1; s \leftarrow s_1; t \leftarrow t_1$ 
```

b. Algorithm

Example

- Given $a = 161$ and $b = 28$, find $\gcd(a, b)$ and the values of s and t .

$$r = r_1 - q \times r_2 \quad s = s_1 - q \times s_2 \quad t = t_1 - q \times t_2$$

We use a table to follow the algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

We get $\gcd(161, 28) = 7$, $s = -1$ and $t = 6$

Example 2

- Given $a = 17$ and $b = 0$, find $\gcd(a, b)$ and the values of s and t .

Solution

We use a table to follow the algorithm.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
	17	0		1	0		0	1	

We get $\gcd(17, 0) = 17$, $s = 1$, and $t = 0$

Set of Residues

- The modulo operation creates a set, which in modular arithmetic is referred to as **the set of least residues modulo n , or Z_n** .
- Some Z_n sets

$$Z_n = \{ 0, 1, 2, 3, \dots, (n-1) \}$$

$$Z_2 = \{ 0, 1 \}$$

$$Z_6 = \{ 0, 1, 2, 3, 4, 5 \}$$

$$Z_{11} = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \}$$

Congruence

- In cryptography, we often used the concept of congruence instead of equality.
- Mapping from $\mathbb{Z}$ to $\mathbb{Z}_n$ is not one-to-one.
- Infinite members of $\mathbb{Z}$ can map to one member of $\mathbb{Z}_n$.
- For example, the result of $2 \bmod 10 = 2$, $12 \bmod 10 = 2$, $22 \bmod 10 = 2$
 - In modular arithmetic, integers like 2, 12, and 22 are called congruent mod 10.

Congruence

- Two numbers r and s are said to be “congruent mod m ” if

$$r \bmod m = s \bmod m$$

- In this case we write $r \equiv s \bmod m$
- The difference between r and s will be a multiple of m

So $r - s = km$ for some value of k

$$2 \equiv 12 \pmod{10}$$

$$13 \equiv 23 \pmod{10}$$

$$3 \equiv 8 \pmod{5}$$

$$8 \equiv 13 \pmod{5}$$

- Congruence operator Vs. equality operator
 - an equality operator maps a member of Z to itself; the congruence operator maps a member from Z to a member of Z_n
 - the equality operator is one-to-one; the congruence operator is many-to-one.
 - the symbol mod in $12 \bmod 10$ is an operator; the phrase (mod 10) in $2 \equiv 12 \pmod{10}$ means that the destination set is Z_{10} .

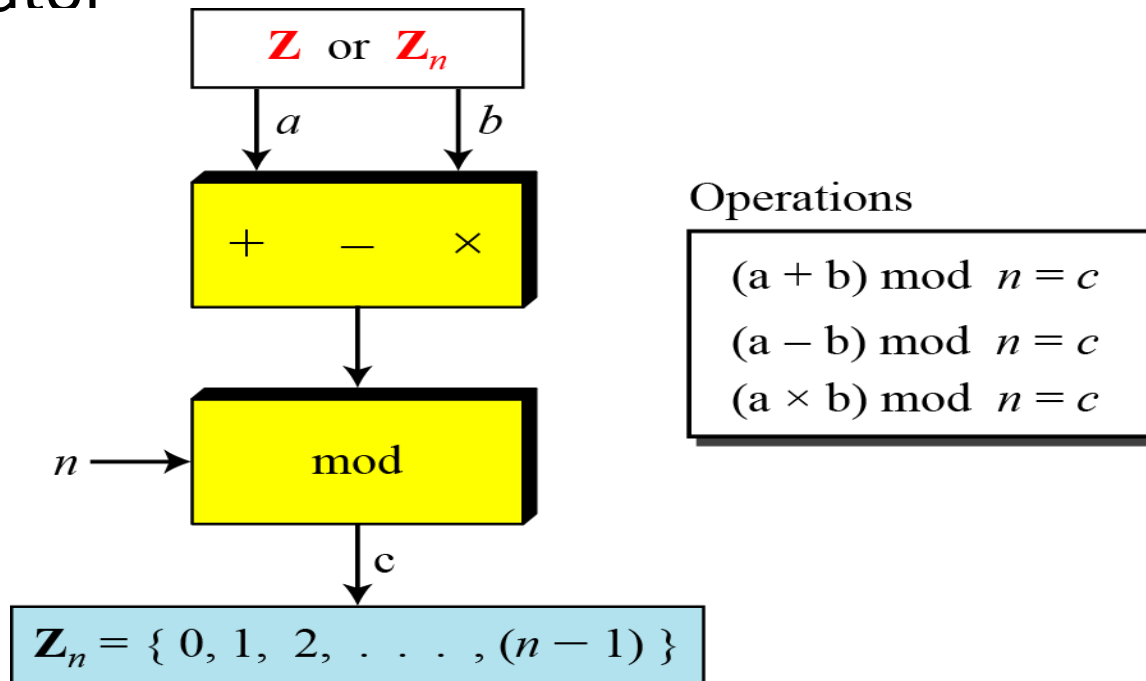
Residue Classes

- A residue class $[a]$ or $[a]_n$ is the set of integers congruent modulo n .
- It is the set of all integers such that $x = a \pmod{n}$.
- For example, if $n = 5$, we have five sets $[0]$, $[1]$, $[2]$, $[3]$, and $[4]$ as shown below:

$$\begin{aligned}[0] &= \{ \dots, -15, -10, -5, 0, 5, 10, 15, \dots \} \\[1] &= \{ \dots, -14, -9, -4, 1, 6, 11, 16, \dots \} \\[2] &= \{ \dots, -13, -8, -3, 2, 7, 12, 17, \dots \} \\[3] &= \{ \dots, -12, -7, -2, 3, 8, 13, 18, \dots \} \\[4] &= \{ \dots, -11, -6, -1, 4, 9, 14, 19, \dots \}\end{aligned}$$

Operation in Z_n

- The three binary operations that applicable for the set Z can also be defined for the set Z_n . The result may need to be mapped to Z_n using the mod operator



Example

Perform the following operations (the inputs come from $\mathbb{Z}_n$):

a. Add 7 to 14 in $\mathbb{Z}_{15}$.

b. Subtract 11 from 7 in $\mathbb{Z}_{13}$.

c. Multiply 11 by 7 in $\mathbb{Z}_{20}$.

Solution

$$(14 + 7) \bmod 15 \rightarrow (21) \bmod 15 = 6$$

$$(7 - 11) \bmod 13 \rightarrow (-4) \bmod 13 = 9$$

$$(7 \times 11) \bmod 20 \rightarrow (77) \bmod 20 = 17$$

Example

Perform the following operations (the inputs come from either $\mathbf{Z}$ or $\mathbf{Z}_n$):

- a. Add 17 to 27 in $\mathbf{Z}_{14}$.
- b. Subtract 43 from 12 in $\mathbf{Z}_{13}$.
- c. Multiply 123 by -10 in $\mathbf{Z}_{19}$.

Solution

The following shows the two steps involved in each case:

$$(17 + 27) \bmod 14 \quad \rightarrow \quad (44) \bmod 14 = 2$$

$$(12 - 43) \bmod 13 \quad \rightarrow \quad (-31) \bmod 13 = 8$$

$$(123 \times (-10)) \bmod 19 \quad \rightarrow \quad (-1230) \bmod 19 = 5$$

Properties

First Property: $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$

Second Property: $(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$

Third Property: $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Examples

$$11 \bmod 8 = 3; 15 \bmod 8 = 7$$

$$[(11 \bmod 8) + (15 \bmod 8)] \bmod 8 = 10 \bmod 8 = 2$$

$$(11 + 15) \bmod 8 = 26 \bmod 8 = 2$$

$$[(11 \bmod 8) - (15 \bmod 8)] \bmod 8 = -4 \bmod 8 = 4$$

$$(11 - 15) \bmod 8 = -4 \bmod 8 = 4$$

$$[(11 \bmod 8) \times (15 \bmod 8)] \bmod 8 = 21 \bmod 8 = 5$$

$$(11 \times 15) \bmod 8 = 165 \bmod 8 = 5$$

A good thing about modular arithmetic is that the numbers you are working with will be kept relatively small. At each stage of an algorithm, the mod function should be applied.

Thus to multiply $39 * 15 \bmod 11$ we first take modulus to get

$$39 \bmod 11 = 6 \text{ and } 15 \bmod 11 = 4$$

The multiplication required is now

$$6 * 4 \bmod 11 = 24 \bmod 11 = 2$$

Example

- We need to find $10 \bmod 7$, $10^2 \bmod 7$, $10^3 \bmod 7$
- The third property of the mod operator is useful to find this.

$$10^n \bmod x = (10 \bmod x)^n \bmod x \quad \text{Applying the third property } n \text{ times.}$$

We have

$$10 \bmod 3 = 1 \quad \rightarrow \quad 10^n \bmod 3 = (10 \bmod 3)^n = 1$$

$$10 \bmod 9 = 1 \quad \rightarrow \quad 10^n \bmod 9 = (10 \bmod 9)^n = 1$$

$$10 \bmod 7 = 3 \quad \rightarrow \quad 10^n \bmod 7 = (10 \bmod 7)^n = 3^n \bmod 7$$

Inverses

- When we are working in modular arithmetic, we often need to find the inverse of a number relative to an operation.
- We are normally looking for an **additive inverse** (relative to an addition operation) or a **multiplicative inverse** (relative to a multiplication operation).

Additive Inverse

- In Z_n , two numbers a and b are additive inverses of each other if

$$a + b \equiv 0 \pmod{n}$$

- In modular arithmetic, each integer has an additive inverse.
- The sum of an integer and its additive inverse is congruent to 0 modulo n .

- In $\mathbb{Z}_n$, the additive inverse of a can be calculated as $b = n - a$.
 - For example, the additive inverse of 4 in $\mathbb{Z}_{10}$ is 10
– 4 = 6.

Example

- Find all additive inverse pairs in $\mathbb{Z}_{10}$.

Solution

- The six pairs of additive inverses are $(0, 0)$, $(1, 9)$, $(2, 8)$, $(3, 7)$, $(4, 6)$, and $(5, 5)$.
 - the additive inverses are **reciprocal**; if 4 is the additive inverse of 6, then 6 is also the additive inverse of 4.

Multiplicative Inverse

- In Z_n , two numbers a and b are the multiplicative inverse of each other if

$$a \times b \equiv 1 \pmod{n}$$

- In modular arithmetic, an integer may or may not have a multiplicative inverse.
- When it does, the product of the integer and its multiplicative inverse is congruent to 1 modulo n .

- The integer a has a multiplicative inverse in $\mathbb{Z}_n$ if and only if $\gcd(n, a) = 1$.
 - In this case, a and n are said to be relatively prime.

Example

- Find the multiplicative inverse of 8 in $\mathbb{Z}_{10}$.

There is no multiplicative inverse because

$$\gcd(10, 8) = 2 \neq 1$$

Example

Find all multiplicative inverses in $\mathbb{Z}_{10}$.

There are only three pairs:

$(1, 1)$, $(3, 7)$ and $(9, 9)$

$$(1 \times 1) \bmod 10 = 1 \quad (3 \times 7) \bmod 10 = 1 \quad (9 \times 9) \bmod 10 = 1$$

Example

Find all multiplicative inverses in $\mathbb{Z}_{11}$.

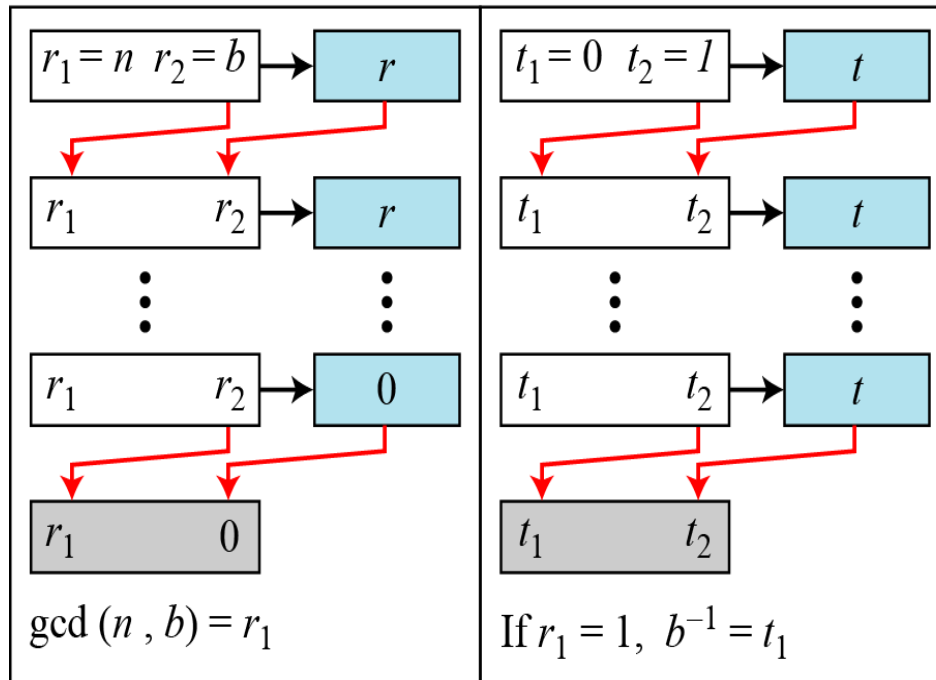
There are seven pairs:

$(1, 1)$, $(2, 6)$, $(3, 4)$, $(5, 9)$, $(7, 8)$, $(9, 9)$, and $(10, 10)$.

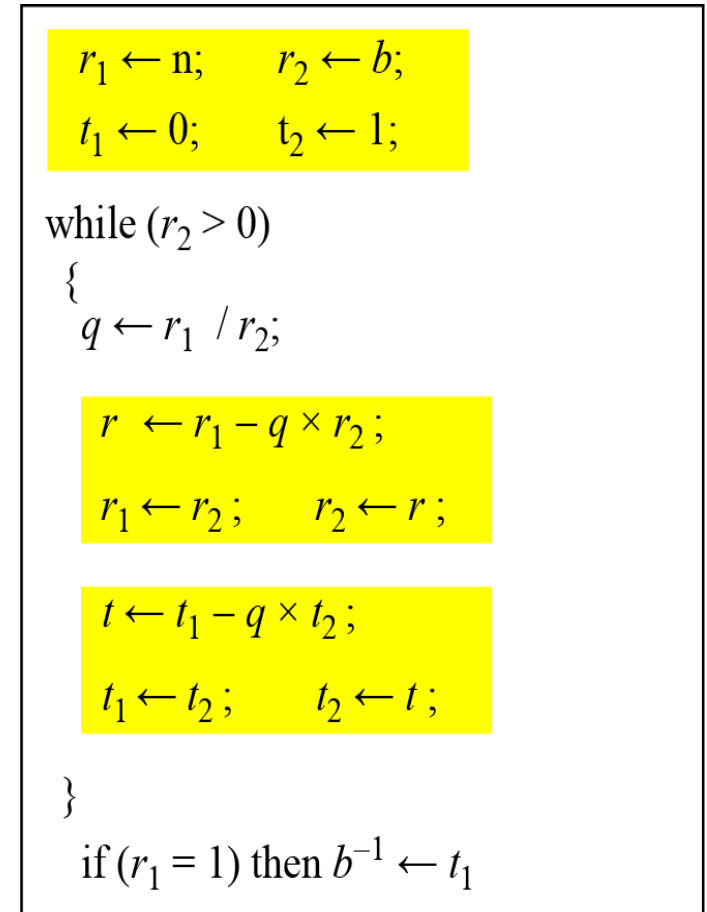
Extended Euclidean algorithm

- The extended Euclidean algorithm finds the multiplicative inverses of b in Z_n when n and b are given and $\gcd(n, b) = 1$.
- The multiplicative inverse of b is the value of t after being mapped to Z_n .
- The algorithm can find s and t such
$$s \times n + b \times t = \gcd(n, b).$$
- The relationship is $(s \times n) + (b \times t) = 1$

Using extended Euclidean algorithm to find multiplicative inverse



a. Process



b. Algorithm

Example

- Find the multiplicative inverse of 11 in $\mathbb{Z}_{26}$.

q	r_1	r_2	r	t_1	t_2	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26
	1	0		-7	26	

The gcd (26, 11) is 1; the inverse of 11 is -7 or 19.

Example

- Find the multiplicative inverse of 23 in $\mathbb{Z}_{100}$.

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	-4
2	23	8	7	1	-4	19
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	0		-13	100	

The gcd (100, 23) is 1; the inverse of 23 is -13 or 87.

Example

- Find the inverse of 12 in $\mathbb{Z}_{26}$.

q	r_1	r_2	r	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

The gcd (26, 12) is 2; the inverse does not exist.

Different Sets

- We need to use $\mathbb{Z}_n$ when additive inverses are needed
- We need to use $\mathbb{Z}_n^*$ when multiplicative inverses are needed.

- Some Z_n and Z_n^* sets

$$Z_6 = \{0, 1, 2, 3, 4, 5\}$$

$$Z_6^* = \{1, 5\}$$

$$Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$$

$$Z_7^* = \{1, 2, 3, 4, 5, 6\}$$

$$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$Z_{10}^* = \{1, 3, 7, 9\}$$

Two More Sets

- Cryptography often uses two more sets: Z_p and Z_p^* .
- The modulus in these two sets is a prime number.

$$Z_{13} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$Z_{13}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Matrices

- A matrix of size $l \times m$

m columns

Matrix **A**:

$$\begin{matrix} \text{\textit{l} rows} \\ \left[\begin{array}{cccc} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{l1} & a_{l2} & \dots & a_{lm} \end{array} \right] \end{matrix}$$

Examples of matrices

$$\begin{bmatrix} 2 & 1 & 5 & 11 \end{bmatrix}$$

Row matrix

$$\begin{bmatrix} 2 \\ 4 \\ 12 \end{bmatrix}$$

Column
matrix

$$\begin{bmatrix} 23 & 14 & 56 \\ 12 & 21 & 18 \\ 10 & 8 & 31 \end{bmatrix}$$

Square
matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

0

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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Operations and Relations

- In linear algebra, one relation (equality) and four operations (addition, subtraction, multiplication, and scalar multiplication) are defined for matrices.

Operations and Relations

- Addition and subtraction of matrices

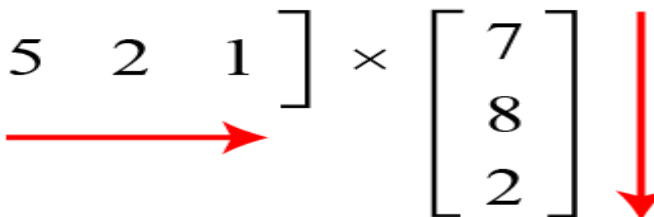
$$\begin{bmatrix} 12 & 4 & 4 \\ 11 & 12 & 30 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 1 \\ 3 & 2 & 10 \end{bmatrix} + \begin{bmatrix} 7 & 2 & 3 \\ 8 & 10 & 20 \end{bmatrix}$$

$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$\begin{bmatrix} -2 & 0 & -2 \\ -5 & -8 & 10 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 1 \\ 3 & 2 & 10 \end{bmatrix} - \begin{bmatrix} 7 & 2 & 3 \\ 8 & 10 & 20 \end{bmatrix}$$

$$\mathbf{D} = \mathbf{A} - \mathbf{B}$$

- Multiplication of a row matrix by a column matrix

$$\begin{array}{c} \mathbf{C} \\ [53] \end{array} = \begin{array}{c} \mathbf{A} \\ [5 \quad 2 \quad 1] \end{array} \times \begin{array}{c} \mathbf{B} \\ \begin{bmatrix} 7 \\ 8 \\ 2 \end{bmatrix} \end{array}$$


In which:

$$53 = 5 \times 7 + 2 \times 8 + 1 \times 2$$

- Multiplication of a 2×3 matrix by a 3×4 matrix

$$\begin{array}{c} \mathbf{C} \\ \left[\begin{array}{cccc} 52 & 18 & 14 & 9 \\ 41 & 21 & 22 & 7 \end{array} \right] \end{array} = \begin{array}{c} \mathbf{A} \\ \left[\begin{array}{ccc} 5 & 2 & 1 \\ 3 & 2 & 4 \end{array} \right] \end{array} \times \begin{array}{c} \mathbf{B} \\ \left[\begin{array}{cccc} 7 & 3 & 2 & 1 \\ 8 & 0 & 0 & 2 \\ 1 & 3 & 4 & 0 \end{array} \right] \end{array}$$

- We can also multiply a matrix by a number (called a scalar).
- Example of scalar multiplication.

$$\begin{matrix} \mathbf{B} \\ \begin{bmatrix} 15 & 6 & 3 \\ 9 & 6 & 12 \end{bmatrix} \end{matrix} = 3 \times \begin{matrix} \mathbf{A} \\ \begin{bmatrix} 5 & 2 & 1 \\ 3 & 2 & 4 \end{bmatrix} \end{matrix}$$

Determinant

The determinant of a square matrix A of size $m \times m$ denoted as $\det(A)$ is a scalar calculated recursively as shown below:

1. If $m = 1$, $\det(\mathbf{A}) = a_{11}$
2. If $m > 1$, $\det(\mathbf{A}) = \sum_{j=1}^m (-1)^{i+j} \times a_{ij} \times \det(\mathbf{A}_{ij})$

Where $\mathbf{A}_{ij}$ is a matrix obtained from $\mathbf{A}$ by deleting the i th row and j th column.

The determinant is defined only for a square matrix.

- Calculating the determinant of a 2×2 matrix

$$\det \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix} = (-1)^{1+1} \times 5 \times \det[4] + (-1)^{1+2} \times 2 \times \det[3] \longrightarrow 5 \times 4 - 2 \times 3 = 14$$

or

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11} \times a_{22} - a_{12} \times a_{21}$$

- Calculation of the determinant of a 3×3 matrix.

$$\det \begin{bmatrix} 5 & 2 & 1 \\ 3 & 0 & -4 \\ 2 & 1 & 6 \end{bmatrix} = (-1)^{1+1} \times 5 \times \det \begin{bmatrix} 0 & -4 \\ 1 & 6 \end{bmatrix} + (-1)^{1+2} \times 2 \times \det \begin{bmatrix} 3 & -4 \\ 2 & 6 \end{bmatrix} + (-1)^{1+3} \times 1 \times \det \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$$

$$= (+1) \times 5 \times (+4) \quad + \quad (-1) \times 2 \times (24) \quad + \quad (+1) \times 1 \times (3) = -25$$

Inverses

- Matrices have both additive and multiplicative inverses.
- The additive inverse of matrix A is another matrix B such that $A + B = 0$.
 - additive inverse of A is defined by $-A$.
- Multiplicative inverses are only defined for square matrices.

- The multiplicative inverse of a square matrix A is a square matrix B such that $A \times B = B \times A = I$.
- Normally the multiplicative inverse of A is defined by A^{-1} .
- The multiplicative inverse exists only if the $\det(A)$ has a multiplicative inverse in the corresponding set.

Residue Matrices

- Cryptography uses residue matrices: matrices where all elements are in $\mathbb{Z}_n$.
- A residue matrix has a multiplicative inverse if $\gcd(\det(A), n) = 1$.

Example

- A residue matrix A in Z_{26} and its multiplicative inverse

$$\mathbf{A} = \begin{bmatrix} 3 & 5 & 7 & 2 \\ 1 & 4 & 7 & 2 \\ 6 & 3 & 9 & 17 \\ 13 & 5 & 4 & 16 \end{bmatrix}$$

$$\det(\mathbf{A}) = 21$$

$$\mathbf{A}^{-1} = \begin{bmatrix} 15 & 21 & 0 & 15 \\ 23 & 9 & 0 & 22 \\ 15 & 16 & 18 & 3 \\ 24 & 7 & 15 & 3 \end{bmatrix}$$

$$\det(\mathbf{A}^{-1}) = 5$$

Congruence

- Two matrices are congruent modulo n , if they have the same number of rows and columns and all corresponding elements are congruent modulo n .
- Written as $A \equiv B \pmod{n}$

$A \equiv B \pmod{n}$ if $a_{ij} \equiv b_{ij} \pmod{n}$ for all i 's and j 's.

Linear Congruence

- Cryptography often involves solving an equation or a set of equations of one or more variables with coefficient in $\mathbb{Z}_n$.
- Also helps to solve equations when the power of each variable is 1 (linear equation).

Single-Variable Linear Equations

- The equations of the form $ax \equiv b \pmod{n}$.
- An equation of this type might have no solution or a limited number of solutions.

Assume that the $\gcd(a, n) = d$.

If $d \nmid b$, there is no solution.

If $d \mid b$, there are d solutions.

If $d|b$, we use the following strategy to find the solutions:

1. Reduce the equation by dividing both sides of the equation (including the modulus) by d .
2. Multiply both sides of the reduced equation by the multiplicative inverse of a to find the particular solution x_0 .
3. The general solutions are $x = x_0 + k (n/d)$ for $k = 0, 1, \dots, (d-1)$.

Example

- Solve the equation $10x \equiv 2 \pmod{15}$.

$$\gcd(10, 15) = 5.$$

Since 5 does not divide 2, there is no solution.

Example

- Solve the equation $14x \equiv 12 \pmod{18}$.

$$\gcd(14 \text{ and } 18) = 2.$$

Since 2 divides 12, it has exactly two solutions

$$\begin{aligned} 14x &\equiv 12 \pmod{18} \rightarrow 7x \equiv 6 \pmod{9} \rightarrow x \equiv 6(7^{-1}) \pmod{9} \\ x_0 &= (6 \times 7^{-1}) \pmod{9} = (6 \times 4) \pmod{9} = 6 \\ x_1 &= x_0 + 1 \times (18/2) = 15 \end{aligned}$$

Both solutions, 6 and 15 satisfy the congruence relation, because $(14 \times 6) \pmod{18} = 12$ and also $(14 \times 15) \pmod{18} = 12$.

Example

- $37x \equiv 15 \pmod{49}$.
- $\gcd(37, 49) = 1$ implies that the equation has exactly ONE solution.
- Find the multiplicative inverse of 37 modulo 49, i.e. $37y \equiv 1 \pmod{49}$.
- Euclidean algorithm: $49 = 37 + 12$, $37 = 3 \cdot 12 + 1$. Thus $1 = 37 - 3 \cdot 12 = 37 - 3(49 - 37) = 4 \cdot 37 - 3 \cdot 49$.
- 4 is the multiplicative inverse of 37 modulo 49.
- $37x \equiv 15 \pmod{49}$, therefore $4 \cdot 37x \equiv 15 \pmod{49}$, i.e. $x \equiv 11 \pmod{49}$.

Example

- $18x \equiv 8 \pmod{14}$.
- $\gcd(14, 18) = 2$ implies that the equation has TWO solutions.
- $2 \cdot 9x \equiv 2 \cdot 4 \pmod{2 \cdot 7}$ is equivalent to $9x \equiv 4 \pmod{7}$ because $\gcd(2, 14) = 2$.
- Find the multiplicative inverse of 9 modulo 7, i.e. $9y \equiv 1 \pmod{7}$.
- Euclidean algorithm: $9 = 7 + 2, 7 = 3 \cdot 2 + 1$. Thus $1 = 4 \cdot 7 - 3 \cdot 9$. -3 is the multiplicative inverse of 9 modulo 7, i.e. $-3 \equiv 4 \pmod{7}$.
- $9x \equiv 4 \pmod{7}$, therefore $4 \cdot 9x \equiv 4 \cdot 4 \pmod{7}$, i.e. $x_0 \equiv 2 \pmod{7}$. The second solution:
 $x_1 = x_0 + 14/2 = 2 + 7 = 9$.

A set of Single-Variable Linear Equations

- We can also solve a set of linear equations with the same modulus if the matrix formed from the coefficients of the variables is invertible.

$$AX = B$$

$$A^{-1}(AX) = A^{-1}B$$

$$(A^{-1}A)X = A^{-1}B$$

$$(I_n)X = A^{-1}B$$

$$X = A^{-1}B$$

$$\begin{array}{rcl}
 a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n & \equiv & b_1 \\
 a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n & \equiv & b_2 \\
 \vdots & & \vdots \\
 a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n & \equiv & b_n
 \end{array}$$

a. Equations

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \equiv \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \equiv \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

b. Interpretation

c. Solution

Assignment

- Solve the set of following three equations:

$$3x + 5y + 7z \equiv 3 \pmod{16}$$

$$x + 4y + 13z \equiv 5 \pmod{16}$$

$$2x + 7y + 3z \equiv 4 \pmod{16}$$

Solution

The result is $x \equiv 15 \pmod{16}$, $y \equiv 4 \pmod{16}$, and $z \equiv 14 \pmod{16}$.